

SINA EBRAHIMI

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PhD Candidate in E2E Network Slicing (expected Jun 2026; available for part-time roles earlier) — CFTC, Coventry University

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SUMMARY

I am a Ph.D. researcher at Coventry University (expected completion: June 2026) specialising in end-to-end (E2E) network slicing for 5G/6G. My work sits at the intersection of mobile network architecture, resource management, and AI-enabled orchestration, informed by prior industry roles in telecom operations and core-network engineering. I am preparing for industry R&D / advanced engineering roles in networking and telecom, with interests spanning network automation/orchestration, NFV/SDN, resource allocation, MEC/task offloading, and deep reinforcement learning for closed-loop network control.

EDUCATION

Ph.D. in E2E Network Slicing

Centre for Future Transport & Cities (CFTC), Coventry University, Coventry, United Kingdom

Dissertation: *5G/6G End-to-End Network Slicing Framework for Vertical Industries* 

Supervisors: Olivier C. L. Haas   , Soufiene Djahel   , Hui Zhou   , and Faouzi Bouali   

October 2022 –


M.Sc. in Software Engineering

Dept. of Electrical & Computer Engineering, Tarbiat Modares University, Tehran, Iran

Thesis: *A Resource Allocation Method for Network Slicing in Software-Defined 5G Networks*  

Supervisors: Behzad Akbari    and Nader Mokari   

September 2017 – January 2020



GPA: 3.54/4.0

B.Sc. in Software Engineering

Dept. of Electrical & Computer Engineering, University of Zanjan, Zanjan, Iran

Capstone: *Design and Implementation of an Appointment Reservation System using ASP.NET MVC* 

Supervisor: Davud Mohammadpur   

January 2012 – November 2016



GPA: 3.31/4.0

PUBLICATIONS

Journal Papers

4. **Ebrahimi, Sina**, Hui Zhou, Soufiene Djahel, Faouzi Bouali, and Olivier C. L. Haas. “ENSURE: Multi-Timescale E2E Slice Resource Management for 5G/6G Networks,” *IEEE Transactions on Network and Service Management* (IF: 5.4), Submitted, pp. 1-12, Jan 2026.  
3. **Ebrahimi, Sina**, Faouzi Bouali, and Olivier C. L. Haas. “Resource Management From Single-domain to End-to-End Network Slicing: A Survey,” *IEEE Communications Surveys & Tutorials* (IF: 46.7), vol. 26, no. 4, pp. 2836-2866, 2024.  
2. Akhtarshenas, Azim, Mohammad Ali Vahedifar, Navid Ayoobi, Behrouz Maham, Tohid Alizadeh, **Sina Ebrahimi**, David López-Pérez. “Federated Learning: A Cutting-Edge Survey of the Latest Advancements and Applications,” *Computer Communications* (IF: 4.3), vol. 228, pp. 107964, 2024.  
1. Tajallifar, Mohsen, **Sina Ebrahimi**, Mohammad Reza Javan, Nader Mokari, and Luca Chiaraviglio. “Energy-Efficient Task Offloading Under E2E Latency Constraints,” *IEEE Transactions on Communications* (IF: 8.3), vol. 70, no. 3, pp. 1711-1725, 2022.  

Conference Papers

3. **Ebrahimi, Sina**, Faouzi Bouali, and Olivier C. L. Haas. “MARSRA: Mobility-Aware RAN Slicing Resource Allocation for Open RAN Deployments,” *IEEE International Workshop on Computer Aided Modeling and Design of Communication Links and Networks*, Athens, Greece, CAMAD, 2024.  
2. Zakeri, Abulfazl, Narges Gholipour, Mohsen Tajallifar, **Sina Ebrahimi**, Nader Mokari, Mohammad Reza Javan, and Ahmad Reza Sharafat. “Digital Transformation VIA 5G: Deployment Plans,” *ITU Kaleidoscope 2020 Conference*, Ha Noi, Vietnam, ITU K, 2020. **Awarded as the third best paper of the conference.* 
1. **Ebrahimi, Sina**, Abulfazl Zakeri, Nader Mokari, and Behzad Akbari. “Joint Resource and Admission Management for Slice-enabled Networks,” *IEEE/IFIP Network Operations and Management Symposium*, Budapest, Hungary, NOMS, 2020.  

WORK EXPERIENCE

Research Assistant

Centre for Future Transport & Cities (CFTC), Coventry University, Coventry, UK

October 2022 –


- Courses taken: [Machine Learning, Spring 2024](#)  and [Artificial Neural Networks, Fall 2024](#) .
- Working/Researching on the simulation and design of an E2E resource management framework for network slicing.

Research Assistant (V2X Systems Integration) — Project CERTUS

January 2025 – July 2025

Centre for Connected & Autonomous Automotive Research (CCAAR), HORIBA MIRA, Nuneaton, UK



- Innovate UK-funded [CERTUS](#): V2X integration/validation on CAV testbeds (supervised by [Dr Haas](#), [Dr Lu](#)).
- Integrated and validated Cohda Wireless [MK6](#) C-V2X units on [StreetDrone](#) CAV platforms.
- Debugged PC5/802.11p (ExampleETSI, Wireshark pcaps) and automated PC-MK6 checks (IPv6/UDP/log scripts).
- Analysed steering logs in Python to estimate control-loop delay and assess deadband mitigation.

Network Solutions Specialist

December 2021 – October 2022

Router R&D Dept., SINA Communication Systems Co., Tehran, Iran

 سامانه های نوآوری ارتباطات سینا

- SINA is a network solution provider and a manufacturer of transmission network devices.
- Tested and verified NOS protocol modules with TRex, GNS3, and Wireshark.
- Experience with OF-DPA, OpenFlow, FRRouting, OpenDaylight, ONOS, and Stratum.

Network Performance Tools Engineer

March 2020 – August 2021

Network Quality & Supervision Dept., RighTel Telecommunications Services Co., Tehran, Iran

 رایتل

- RighTel is the third 3G/4G telecom operator in Iran.
- Built KPI ETL + Grafana dashboards for OSS telemetry; automated daily network health reporting and downtime alerts.
- Installation and maintenance of local PostgreSQL and Gitlab CE servers on Ubuntu VMs.

5G Core Network Researcher

January 2018 – July 2019

Mobile Network Technology Solution & Strategy Dept., Mobile Communication Company of Iran (MCI), Tehran, Iran

 مخابرات ایران

- A collaborative project conducted by Dr. Mokari's team at Tarbiat Modares University.
- Topics covered by me include *5G Mobile Core Network Architecture*, *SDN*, *NFV*, *Orchestration*, *Network Slicing*, *5G Core Network Functions*, and *5G System Design*.
- Collaborated with team members to prepare reports, bulletins, and presentations, focusing on studies of technologies, standardization, and migration plans.
- Commended by the MCI Technical Deputy for outstanding research on 5G technologies and standards.

Research/Teaching Assistant

January 2018 – January 2020

Wireless Innovation Lab, Dept. of Electrical & Computer Engineering, Tarbiat Modares University, Tehran, Iran



- Published three papers and enhanced my teamwork and documentation skills.
- Helping newly-arrived M.Sc. students with their research.
- Teaching Assistant in Advanced Computer Networks (Lecturer: Dr. Behzad Akbari, M.Sc. students, Fall 2018).

SKILLS

Core: 5G/6G, Network Slicing, NFV, SDN, O-RAN, MEC, Telecom Standards, Resource Allocation, Closed-loop Automation

Networking: Wireshark, TRex, GNS3, OpenFlow, FRRouting, ONOS, Stratum

Programming/Data: Python, SQL (PostgreSQL/MS SQL), .NET Core, MATLAB, pandas, NumPy, PyTorch, scikit-learn

Research: $\LaTeX$, MS Office, EndNote, GenAI Usage, DRL (Stable-Baselines3, Ray RLLib), Gymnasium, CVX, Gurobi

Tools: Linux, Git, GitLab, Grafana, Docker

CERTIFICATES

 IELTS, IDP, April 21, 2022. (Overall Score: 7.5; L: 9, R: 7.5, W: 6.5 S: 7.5)

 Machine Learning, Stanford (Coursera), Online, Summer 2021

 Mobile Broadband Internet, 5G and Future Services, ITU Academy, Online, Spring 2021

 LPIC-2 & LPIC-1, Fanavaran Anisa (Iran Linux House), Tehran, Iran, Spring 2021 & Winter 2018

 CCNA, Carrier Digit Institute, Tehran, Iran, Summer 2017

ACADEMIC AND PROFESSIONAL MEMBERSHIP

January 2020 - IEEE Graduate Student Member (No.: 95013754) 
 August 2024 - Member of IEEE ComSoc, VTS, CS, YP, and R8—UK & Ireland Section 
 August 2024 - October 2024 TPC member of the 2024 IEEE 100th Vehicular Technology Conference (VTC2024-Fall) 

VOLUNTEERING

Graduate Student Mentor (Volunteer)

January 2024 -

- Assisted 10+ candidates with PhD applications, leading to 7 successful placements in 2024.

Board Member, Simorgh Film & Photo Center , University of Zanjan , Zanjan, Iran

October 2014 - October 2016

- Organized screenings, subtitled films, produced videos, and managed board elections to promote arts at the university.
- Directed ticket proceeds as donations to the Iranian Society of Students Against Poverty (SOSA Poverty).

REVIEW/TPC SERVICES

Journals:  IEEE Transactions on Mobile Computing (TMC) (IF: 9.2) , IEEE Open Journal of the Communications Society (OJ-COMS) (IF: 6.1) , IEEE Networking Letters (LNET) (IF: 2.7) 
 EURASIP Journal on Wireless Communications and Networking (JWCN) (IF: 1.9)  

Conferences: IEEE Vehicular Technology Conference (VTC2024-Fall), Washington DC, United States.  ; IEEE International Conference on Communications (ICC 2025), Montreal, Canada.  ; IEEE Wireless Communications and Networking Conference (WCNC 2025), Milan, Italy.  ; IEEE Mediterranean Communication and Computer Networking Conference (MedComNet 2025), Cagliari, Italy. ; IEEE International Conference on Computer Communications (INFOCOM 2025), London, United Kingdom.  

EVENTS, KEYNOTES, AND WORKSHOPS

Presented:

- November 2024: DRL-enabled Mobility-Aware RAN Slicing Resource Allocation for Open RAN Deployments, [Advanced Internet Technologies: Winter PhD School 2024](#),  Barcelona, Spain (Travel costs covered by the Spanish Government's UNICO I+D 6G  and Teleco Renta  initiatives).
- November 2024: Conversations with Graduates Webinar Series: Sina Ebrahimi ([Online Interview](#)),  MFet Webinar Series (In coordination with IEEE Sharif UT Student Branch), ZNU Electrical Engineering Scientific Association , Zanjan, Iran.
- October 2024: Mobility-Aware RAN Slicing Resource Allocation for Open RAN Deployments ([Session 2: Distributed AI-Driven Open & Programmable Architecture for 6G Networks](#)),  IEEE CAMAD 2024,  Athens, Greece.
- September 2024: Application of Deep Reinforcement Learning to RAN Slicing ([Online Presentation](#)), Network Softwarization and Security Labs (NetsLab) , School of Computer Science, University College Dublin,  Dublin, Republic of Ireland.
- April 2020: Joint Resource and Admission Management for Slice-enabled Networks ([Session 2: SDN & NFV - Resource & Slice Ma](#)  IEEE/IFIP NOMS 2020,  Budapest, Hungary.
- February 2020: An Introduction to SDN, NFV, and Network Slicing, Niroo Research Institute (NRI),  Tehran, Iran.
- May 2019: REST APIs in the 3GPP 5G Standards, Mobile Network Technology Solution & Strategy Dept., Mobile Communication Company of Iran (MCI),  Tehran, Iran.
- June 2019: 5G Core Network Migration (from EPC), Mobile Network Technology Solution & Strategy Dept., Mobile Communication Company of Iran (MCI),  Tehran, Iran.
- November 2018: 5G Standardization: NFV, SDN, Network Slicing, E2E Architecture, Service-Based Architecture, Mobile Network Technology Solution & Strategy Dept., Mobile Communication Company of Iran (MCI),  Tehran, Iran.
- October 2018: 5G Core Network Overview, 2018 Iran Telecom Exhibition,  Tehran, Iran.
- May 2018: Recent Trends in 5G Core Network Technologies: IMT-2020 Services and Requirements, NFV, SDN, Network Slicing, 5G-PPP and EU Horizon 2020 Projects, 3GPP 5G Deployment Options (SA/NSA), Mobile Network Technology Solution & Strategy Dept., Mobile Communication Company of Iran (MCI),  Tehran, Iran.

Attended:

- September 2025: [Connected Britain 2025: The Future of UK Connectivity](#),  ExCeL, London, UK.

- October 2024: [Human-Centred AI: An Interdisciplinary Network](#),  Coventry, UK.
- March 2024: [Advanced Digital Technologies International Student Workshop](#),  Barcelona, Spain (*Travel costs covered by the Spanish Government's UNICO I+D 6G  and Teleco Renta  initiatives*).
- May 2018: Nokia 5G Update for MCI Workshop, [NOKIA](#)  Tehran, Iran.

REFERENCES

1. **Soufiene Djahel**, Professor in CAVs, Centre for Future Transport & Cities (CFTC), Coventry University,    ae3095@coventry.ac.uk
2. **Olivier C. L. Haas**, Associate Professor and Reader in Applied Control Systems, Centre for Future Transport & Cities (CFTC), Coventry University,    o.haas@coventry.ac.uk
3. **Faouzi Bouali**, Assistant Professor in Telecom & Network Engineering, College of Engineering & Technology, University of Doha for Science & Technology (UDST),    faouzi.bouali@udst.edu.qa
4. **Behzad Akbari**, Deputy Minister and CEO, Iran Telecommunication Infrastructure Company (TIC); Associate Professor, Dept. of Electrical & Computer Engineering, Tarbiat Modares University,    b.akbari@modares.ac.ir
5. **Nader Mokari**, Associate Professor, Dept. of Electrical & Computer Engineering, Tarbiat Modares University,    nader.mokari@modares.ac.ir
6. **Amir Reza Azmoode**, Router Software Team Lead, Router R&D Dept., SINA Communication Systems Co., Tehran, Iran,   a.azmoode@sinacomsys.com
7. **Mehdi Moazzamipeiro**, Network and Service Architecture Manager, Mobile Network Technology Solution & Strategy Dept., Mobile Communication Company of Iran (MCI), Tehran, Iran,   mm.peiro@mci.ir
8. **Mohammad Reza Ahmadi**, Network Quality & Supervision General Manager, Technical Dept., RighTel Telecommunications Services Co., Tehran, Iran,   m.ahmadi@rightel.ir